

Introduction to Systems Neuroscience | Tutorial Notes

Auditory System

MAP TO NEUROSCIENCE CORE CONCEPTS¹

- NCC1. The brain is the body's most complex organ.
- NCC3. Genetically determined circuits are the foundation of the nervous system.
- NCC6. The brain makes it possible to communicate knowledge through language.
- NCC7. The human brain endows us with a natural curiosity to understand how the world works.

KEY CONCEPTS²

- 10.1 Sound is a pressure wave composed of different frequencies important for speech, music, and other natural sounds.
- 10.2 The structures of the ear filter the frequencies of sound and transmit vibrations from air to fluid.
- 10.3 Sound waves are transduced into neural signals by cochlear hair cells.
- 10.4 Transduction is controlled by active mechanisms involving middle ear muscles and cochlear outer hair cells.
- 10.5 Auditory pathways involve signals traveling bilaterally and in both feedforward and feedback directions.
- 10.6 Auditory perception involves the synthesis of multiple aspects of sound.
- 10.7 Neural codes for sound frequency are based on resonance and synchrony.
- 10.8 Sounds are localized based on frequency, relative loudness, and relative time of arrival at the ears.
- 10.9 Hearing coordinates with vision via numerous signal transformations and robust intersensory cross-talk.

TUTORIAL OUTLINE

- I. Introduction to the Auditory System
 - A. overview of auditory function: *the auditory system transduces sound waves into distinct patterns of neural activity that are then integrated with other sensations and motivations to guide behavior*

¹ Visit [BrainFacts.org](https://www.brainfacts.org) for Neuroscience Core Concepts (©2012 Society for Neuroscience) that offer fundamental principles about the brain and nervous system, the most complex living structure known in the universe.

² Key Concepts are aligned to *Neuroscience, 7th Ed.*, Oxford University Press, Inc., 2023. [[click here](#)]

1. sound waves are collected and amplified by physical structures in the external and middle ear for transfer to neural elements in the inner ear
2. the biomechanical properties of the inner ear decompose complex sound waves into sinusoidal components, so that component frequencies, their amplitude, and their phase can be encoded in the firing of the receptor cells
3. in the inner ear and throughout central processing stations, **tonotopy** is preserved; i.e., the systematic representation of sound frequency
4. in the brainstem, the auditory information is first processed and divided into several parallel pathways, some of which compare the signals derived from the two ears, a process that allows for the localization of sounds in space
5. in addition to local processing of auditory signals, brainstem centers relay information to the midbrain, which in turn projects to the auditory thalamus
6. the auditory cortex then receives input from the thalamus and processes more complex aspects of sounds, such as those related to human speech
7. at each stage of central auditory processing, there are extensive connections from “higher” centers to antecedent processing centers, and well as connections that integrate signals arriving at the two ears
8. like all other sensory systems, hearing employs a variety of active mechanisms by which the brain can shape the incoming signals

CONCEPT 10.1 Sound is a pressure wave composed of different frequencies important for speech, music, and other natural sounds.

Learning Objectives:

- 10.1.1 Explain what sound is and how its frequency content is analyzed scientifically.
- 10.1.2 Define pitch, loudness, and timbre.
- 10.1.3 Discuss the range of intensities and frequencies of sound that humans can hear, and how it compares with that of other species.

II. Sound

- A. spherical pressure waves generated by vibrating air molecules (see **Figure 10.1³**) that can be characterized by their amplitude (loudness), frequency (pitch) and phase (temporal displacement)
- B. most natural sounds (human speech, for example) are acoustically complex (see **Figure 10.2** and **10.3**)
- C. humans can detect sound in the frequency range of **20 Hz to 20 kHz**

³ Figure references to *Neuroscience*, 7th Ed., Oxford University Press, Inc., 2023. [[click here](#)]

CONCEPT 10.2 *The structures of the ear filter the frequencies of sound and transmit vibrations from air to fluid.*

Learning Objectives:

- 10.2.1 Identify the structures of the ear and their respective roles with regard to sound frequency, sound amplitude, and transmission from air to a fluid medium.
- 10.2.2 Explain how sounds of different frequencies activate different regions within the cochlea.

III. The ear

- A. external ear (pinna, concha, auditory meatus) (see [Figure 10.4](#))
 - 1. gathers sound energy and focuses it on the **tympanic membrane** (ear drum)
 - 2. filters different sound frequencies to provide cues about sound localization
- B. middle ear (tympanic membrane, ossicles)
 - 1. transmit acoustic energy from air to the inner ear
 - 2. amplifies the pressure of acoustic energy some 200-fold
- C. inner ear (cochlea, auditory nerve)
 - 1. two-fold function:
 - a. biomechanical: decompose complex acoustic energy into component sinusoidal waveforms
 - b. neural (sensory transduction): transduce this mechanical energy into neural signals that are then communicated to the brain
 - 2. **cochlea** (Greek, *cochlos* = snail)
 - a. basic anatomy (see [Figure 10.5](#))
 - (i) small, coiled structure surrounded by bone
 - (ii) bisected by the cochlear partition into two, large fluid filled channels (scala vestibuli, scala tympani)
 - (iii) the cochlear partition contains the **organ of Corti**, which consists of the **basilar** and **tectorial membranes**
 - (iv) the basilar membrane supports inner and outer hair cells; the **inner hair cells** are the sensory receptor cells of the cochlea
 - (v) the cochlear partition contains a smaller, fluid filled channel (scala media), which is important for maintaining the proper ionic environment for neural signaling
 - (vi) the cochlear partition terminates before the apical end of the cochlea, allowing for continuity between the scala vestibuli and scala tympani at the helicotrema
 - b. functional overview

- (i) pressure transmitted by the ossicles of the middle ear causes the **oval window** to bulge inward
 - (ii) this pressure is conducted throughout the fluid of the scala vestibuli (called perilymph), around the helicotrema and into the scala tympani, and is relieved by the outward bulging of the **round window**
 - (iii) passage of pressure throughout the channels of the cochlea sets up vibrations in the basilar membrane (see [Figure 10.6](#))
 - (iv) movements of the basilar membrane deform the ends of the inner hair cells leading to the transduction of mechanical energy into neural impulses (see [Figure 10.7](#))
- c. basilar membrane
- (i) frequency tuning (refer back to [Figure 10.6](#))
 - the membrane is stiffer, thicker and narrower at the base of the cochlea and wider, thinner and more flexible at the apex of the cochlea
 - thus, the base is tuned for (responds best to) high frequencies, and the apex is tuned for low frequencies
 - complex sounds cause vibrations of different parts of the basilar membrane
 - tuning is sharpened by an active process involving the outer hair cells

CONCEPT 10.3 *Sound waves are transduced into neural signals by cochlear hair cells.*

Learning Objectives:

- 10.3.1 Recognize inner hair cells as the source of auditory sensory input to the brain.
- 10.3.2 Explain the role of stereocilia and tip links in the transduction of sound waves into electrical signals.
- 10.3.3 Describe the patterns of ionic flow that occur during transduction of sound waves into electrical signals in inner hair cells.

IV. Sensory transduction

- A. sensory transduction involves continuous fluxes in potassium currents at the tips of hair cell stereocilia (see [Figure 10.8-10.9](#))
 - 1. the stereocilia of hair cells protrude into the scala media, which contains **endolymph** – a solution that contains high concentrations of K^+
 - 2. vibration of the basilar membrane causes a shearing motion between the basilar and tectorial membranes

3. motion bends the **stereocilia** that protrude from the apical ends of the **inner hairs cells**
4. deformation of the stereocilia *toward* the longest stereocilia leads to inward flow of current through K^+ channels and *depolarization* of the hair cell membrane
5. depolarized hair cells release more neurotransmitter on afferent endings, resulting in an increase in action potential firing in the auditory nerve
6. deformation of the stereocilia *away from* the longest stereocilia leads to closure of K^+ channels and *hyperpolarization* of the hair cell membrane
7. hyperpolarized hair cells release less neurotransmitter on afferent endings, resulting in a decrease in action potential firing in the auditory nerve

V. Sensory coding in the auditory nerve

- A. temporal code: hair cell receptor potentials (and action potentials in the auditory nerve) can follow an acoustical stimulus up to about 3 kHz (see [Figure 10.10](#))
- B. “labeled-line” code: preserves information about frequency since the afferent fibers that innervate different parts of the basilar membrane inherit their tuning (from their location of innervation along the length of the basilar membrane)

CONCEPT 10.4 *Transduction is controlled by active mechanisms involving middle ear muscles and cochlear outer hair cells.*

Learning Objectives:

- 10.4.1 Explain why active forces are needed to account for the responsiveness of the living cochlea to sound.
- 10.4.2 Describe the role of outer hair cells in generating such active forces.

VI. Active processes in the cochlea contribute to hearing

- A. evidence for active processing in the cochlea
 1. the tuning curves measured in the auditory nerve are too sharp to be the simple product of passive mechanisms
 2. at very low sound pressure levels, the basilar membrane vibrates much more than would be expected based on its response to high sound pressure levels
 3. the ear can generate sounds: otoacoustic emissions (see [Figure 10.11](#))
- B. the role of outer hair cells
 1. outer hair cells have a motoric function whereby their cell bodies can expand and contract
 - a. outer hair cells receive efferent innervation from the auditory brainstem through the vestibulocochlear nerve (CN VIII)

- b. thus, CN VIII is not purely sensory in a strict sense
- 2. since the stereocilia of outer hair cells are embedded within the tectorial membrane, the expansion and contraction of outer hair cell bodies provides a means for modulating the sheering forces between the basilar and tectorial membranes
- 3. such modulation can alter the local tuning properties of the how the basilar membrane responds to fluid dynamics in the scala vestibuli and scala tympani
- 4. this is thought to be important in amplifying the response of the basilar membrane to very low sound pressure levels
- C. the role of middle ear muscles
 - 1. tensor tympani and stapedius dampen the movements of the maleus and stapedius, respectively (see [Figure 10.4](#))
 - 2. this is thought to be important in dampening down the transmission of forces to the inner ear produced by very high sound pressure levels
 - 3. may also decrease responsiveness to anticipated sounds, such as one's own speech
 - 4. may also provide a mechanism for tuning the auditory system to sound localized to anticipated visual targets that are the source of sound (linking the auditory and visual systems)
 - a. fun fact: your tympanic membranes move when you move your eyes!

STUDY QUESTIONS

- Q1. Which of the following is the primary function of the three bones in the middle ear?
- A. Selective transmission of high-frequency sounds
 - B. Selective transmission of low-frequency sounds
 - C. Amplification of sound pressure waves to increase auditory sensitivity
 - D. Dampening sound pressure waves to prevent damage to the ear
 - E. Facilitation of fluid drainage from the Eustachian tube
- Q2. Which of the following statements about sensory transduction by hair cells is most accurate?
- A. Bending of the cilia toward the shortest cilium produces depolarization.
 - B. Bending of the cilia toward the longest cilium produces hyperpolarization.
 - C. Neurotransmitter is released by hair cells when the calcium that rushes into the cilia reaches the active zone at the base of the hair cells.
 - D. The firing of action potentials in second-order sensory neurons can be either up- or down-regulated, depending on the direction in which the bundle of cilia (of the afferent hair cell) is bent.
 - E. Hair cells are postsynaptic to second-order sensory neurons.

- Q3. What explains the tonotopy that is present in the auditory division of CN VIII?
- A. The amount of myelin wrapping any given axon: the more myelin, the higher the pitch encoded by the axon.
 - B. The location of the cell body in the spiral ganglion: the closer to the center of spiral, the lower the pitch encoded by the axon.
 - C. The location of where the peripheral process receives contact from hair cells along the basilar membrane.
 - D. The average rate of action potential generation: the higher the average rate of action potential generation, the higher the pitch encoded by the axon.
 - E. The number of AMPA receptors in the postsynaptic process: the more AMPA receptors, the higher the pitch encoded by the axon.

LEARNING OBJECTIVES FOR CONCEPTS 10.5-10.9

- 10.5.1 Identify the major areas of the ascending and descending auditory pathways.
- 10.6.1 Discuss different aspects of sound perception and how they interact with one another.
- 10.7.1 Describe the two types of neural representation of sound frequency.
- 10.7.2 Discuss some of the key puzzles in how these representations account for perception.
- 10.8.1 Name the three cues used to localize sound and explain what they entail.
- 10.9.1 Explain what is meant by the differences in reference frame and coding format between the auditory and visual systems.

VII. Central processing of auditory information

- A. subcortical pathways (see [Figure 10.12](#))
 - 1. first order neurons: spiral ganglion cells (bipolar neurons)
 - a. peripheral process innervates inner hair cells
 - b. central process enters the pontine-medullary junction and bifurcates to innervate one or more divisions of the **cochlear nuclei**
 - 2. second order neurons: cells in the cochlear nucleus that project to multiple targets on both sides of the brainstem
 - a. (contralateral) **nucleus of the lateral lemniscus** in the upper pons: involved in detecting presence and temporal properties of sound from one ear (a monaural pathway)
 - a. (bilateral) **superior olivary complex** in the mid-pons: different divisions of this complex are involved in localizing the sources of sounds in auditory space
 - (i) **medial superior olive (MSO)** (see [Figure 10.17](#)): localizes low frequency sounds based on interaural timing differences (much more developed of the two superior olivary nuclei in humans)

- (ii) **lateral superior olive (LSO)** (see [Figure 10.18](#)): localizes sound based on interaural intensity differences (vestigial in humans)
- 3. inferior colliculus
 - a. all lower auditory projections converge on the **inferior colliculus**
 - b. here, a complete map of auditory space is computed
- 4. auditory thalamus
 - a. inferior colliculus projects to the **medial geniculate complex (MGC)**
 - b. cells in the MGC are sensitive to particular combinations of sounds with distinct spectral and temporal characteristics
- B. auditory cortex
 - 1. target of the MGC, i.e., the primary auditory cortex, is located on the superior aspect of the temporal lobe
 - 2. contains several subdivisions (see [Figure 10.13](#))
 - a. **“core”** area or **primary auditory cortex** that receives tonotopic input from the MGC; also maps binaural interactions
 - b. **“belt”** of additional, higher-order auditory areas
 - 3. asymmetry in structure and function
 - a. the posterior portion of the auditory belt contains a network of neural connections that is specialized for **language reception and speech comprehension**; also known as **Wernicke’s area**
 - b. for most people (>99% of right-handers and >90% left-handers), functional language reception and speech comprehension is in the left hemisphere
 - c. there is a structural asymmetry associated with this functional asymmetry: the *planum temporale* (the superior plane of the temporal lobe)
 - i. the left planum temporale (“temporal plane”) is larger in most humans than the right planum temporale
 - ii. the degree of asymmetry is associated with certain specialized auditory functions, such as “perfect pitch” (greater asymmetry in people with perfect pitch)
 - d. functional asymmetry:
 - i. activation of the right hemisphere is typically greater than the left when listening to music
 - ii. activation of the left hemisphere is typically greater when listening to speech and environmental sounds
 - iii. however, both hemispheres participate in vary degrees to processing all aspects of auditory information

VIII. Challenges in intersensory integration

- A. visual cues can influence auditory perception (consider the “McGurk Effect”)
- B. coding sound localization relative to vision (see [Figure 10.19](#))
 - 1. sound localization in the auditory system is based on head-centered coordinates (obviously, the ears are fixed); but the eyes can move
 - 2. visual locations are encoded in visuotopic (retinotopic) coordinates
 - 3. since the eyes have ~40 degrees of rotational mobility in the orbits, visuotopic and head-centered coordinates are often displaced
 - 4. implies that the brain must account for eye movements when localizing sources of sounds in a visual scene
- B. visual receptive fields are small and bounded; but auditory receptive fields are large (as large as a hemifield of auditory space)
- C. it is not yet understood how intersensory integration is achieved for localizing the sources of sounds—a major challenge of systems neuroscientists in this field; however, it is likely that both subcortical (superior and inferior colliculi) and cortical processing (temporal and parietal associational cortex) is involved

STUDY QUESTIONS

- Q4. Which of the following most depends upon the utilization of **bilateral auditory information**?
- A. frequency discrimination
 - B. sound localization
 - C. distinguishing pitch from timbre
 - D. encoding of speech sounds
 - E. detection of very faint sounds
- Q5. Which auditory structure first displays pronounced selectivity for particular combinations of sounds with distinct spectral and temporal characteristics in the auditory pathway?
- A. cochlear nucleus
 - B. lateral superior olive
 - C. medial superior olive
 - D. nuclei of the lateral lemniscus
 - E. inferior colliculus
 - F. medial geniculate complex
 - G. auditory cortex

Answers to Study Questions: #1 = C; #2 = D; #3 = C; #4 = B; #5 = F